

DELFT UNIVERSITY OF TECHNOLOGY
Faculty of Electrical Engineering, Mathematics and Computer Science



Homework Unit 4&5 (ET4260) 2026
Semiconductor Sensors &
Introduction to Finite Element Modelling

Instructions. Document your calculations properly with derivations and screenshots when needed. Submission in Word or PDF format.

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Part A: Hall Sensors

Task 1 a) Best intrinsic semiconductor. The current-related sensitivity of a Hall plate is $S_I = rG/(nqt)$ and the voltage-related sensitivity is $S_V = \mu rG/(\frac{L}{W})_{\text{eff}}$. For an intrinsic material $n = n_i$, so a good Hall sensor wants a *low* intrinsic carrier concentration and a *high* mobility. Reading Table 1:

	Si	GaAs	Ge
n_i [1/cm ³]	1.5×10^{10}	1.8×10^6	2.4×10^{13}
μ_n [cm ² /(V s)]	1350	8500	3900

GaAs wins on *both* counts: its intrinsic concentration is four orders of magnitude below Si (boosting $S_I \propto 1/n_i$) and its electron mobility is the highest of the three (boosting $S_V \propto \mu$). Its wide bandgap (1.42 eV) also keeps thermally generated carriers low, so the response stays stable with temperature. **GaAs is the most suitable.**

Task 1 b) Parameter linking S_I and S_V . Since $V = I \cdot R$, the two sensitivities are tied through the **device resistance R** :

$$S_V = \frac{S}{V} = \frac{S}{IR} = \frac{S_I}{R}, \quad R = R_{\square} \left(\frac{L}{W}\right)_{\text{eff}}.$$

R matters in its own right because it fixes the bias power $P = V^2/R = I^2 R$ (hence self-heating) and the thermal-noise floor $u_n^2 = 4k_B T R \Delta f$. Designing only for sensitivity while ignoring R can push the device into excessive power dissipation or poor SNR.

Task 1 c) Figure of merit for integrated devices. An integrated Hall device runs from a fixed supply *voltage* and a limited power budget, so the **voltage-related sensitivity S_V** (V/(V T)) is the meaningful figure of merit, since it is the sensitivity per unit bias and ultimately sets the achievable SNR-per-power, $\text{SNR}_P \propto \mu rG/(\frac{L}{W})_{\text{eff}}$.

Task 1 d) GaN Hall devices. For the Gen 2 device $S_I = 90 \text{ V}/(\text{A T})$, $S_V = 57 \text{ mV}/(\text{V T})$, $R_{\square} = 430 \Omega/\square$, with $r = 1$.

(i) From $S_V = S_I/(R_{\square} (\frac{L}{W})_{\text{eff}})$,

$$\left(\frac{L}{W}\right)_{\text{eff}} = \frac{S_I}{S_V R_{\square}} = \frac{90}{0.057 \times 430} = \mathbf{3.67}.$$

The geometry factor follows from

$$G \approx \frac{\left(\frac{L}{W}\right)_{\text{eff}}^2}{\sqrt{\left(\frac{L}{W}\right)_{\text{eff}}^4 + \frac{1}{2} \left(\frac{L}{W}\right)_{\text{eff}}^2 + 4}} = \frac{3.67^2}{\sqrt{3.67^4 + \frac{1}{2} 3.67^2 + 4}} = \mathbf{0.97}.$$

(G is close to unity, as expected for the relatively large L/W of a cross/Greek-cross plate.)

(ii) Gen 3 shares Gen 2's geometry, so $(\frac{L}{W})_{\text{eff}} = 3.67$ and $G = 0.97$ carry over. With $S_I = 34$, $S_V = 34 \text{ mV}/(\text{V T})$,

$$R_{\square, \text{Gen3}} = \frac{S_I}{S_V (\frac{L}{W})_{\text{eff}}} = \frac{34}{0.034 \times 3.67} = \mathbf{272 \Omega/\square}.$$

(iii) Sheet carrier density from $S_I = rG/(n_s q)$ and mobility from $S_V = \mu rG/(\frac{L}{W})_{\text{eff}}$:

$$n_s = \frac{rG}{S_I q}, \quad \mu = \frac{S_V (\frac{L}{W})_{\text{eff}}}{rG}.$$

	$n_s = n \cdot t$ [1/cm ²]	μ [cm ² /(V s)]	cross-check $\mu = 1/(R_{\square} q n_s)$
Gen 2	6.74×10^{12}	2154	2154 ✓
Gen 3	1.78×10^{13}	1285	1285 ✓

Gen 3's InAlN barrier gives a roughly $2.6\times$ higher 2DEG density but a lower mobility, the usual density/mobility trade-off in nitride heterostructures.

(iv) Additional: silicon comparison ($(\frac{L}{W})_{\text{eff}} = \sqrt{2}$, so $G = 2/\sqrt{4+1+4} = 2/3 = 0.67$; $S_I = 300$, $S_V = 60 \text{ mV}/(\text{V T})$):

$$R_{\square, \text{Si}} = \frac{300}{0.060\sqrt{2}} = \mathbf{3536} \ \Omega/\square, \quad n_s = \frac{0.67}{300q} = 1.39 \times 10^{12} \text{ 1/cm}^2,$$

$$\mu_{\text{Si}} = \frac{0.060\sqrt{2}}{0.67} = \mathbf{1273} \text{ cm}^2/(\text{V s}).$$

The Si plate has $\sim 8\times$ the sheet resistance of GaN Gen 2 and only $\sim 60\%$ of its mobility. GaN therefore delivers both a lower $R_{\square}$ (more current at fixed power) and a higher mobility, which is exactly why it makes the better integrated Hall material.

Part B: Temperature Sensors

Task 3 a) Bias-related sensitivity of the bridge. With V_B at the top, the left arm is R_1 (top) + R_3 (bottom) and the right arm is $R_2 + R_4$; the output is $V_o = V_A - V_C$ across the two mid-nodes. The figure pairs opposite-TCR resistors in adjacent arms (R_1, R_4 negative; R_2, R_3 positive), which is the fully-active configuration. Writing $R_n = R(1 + \alpha_{\text{neg}}\Delta T)$ and $R_p = R(1 + \alpha_{\text{pos}}\Delta T)$,

$$V_o = V_B \left[\frac{R_3}{R_1 + R_3} - \frac{R_4}{R_2 + R_4} \right] = V_B \frac{(\alpha_{\text{pos}} - \alpha_{\text{neg}}) \Delta T}{2 + (\alpha_{\text{pos}} + \alpha_{\text{neg}}) \Delta T}.$$

The bias-related sensitivity is the output per unit bias per kelvin, evaluated at balance ($\Delta T \rightarrow 0$):

$$S_{\text{bias}} = \frac{1}{V_B} \frac{\partial V_o}{\partial T} \Big|_0 = \frac{\alpha_{\text{pos}} - \alpha_{\text{neg}}}{2} = \frac{5 \times 10^{-4} - (-3 \times 10^{-4})}{2} = \mathbf{4 \times 10^{-4} \text{ 1/K}} (= 0.4 \text{ mV}/(\text{V K})).$$

Task 3 b) SNR vs. temperature. Signal: the output grows essentially linearly with ΔT , $V_o \approx V_B(\alpha_{\text{pos}} - \alpha_{\text{neg}})\Delta T/2$, and is *zero* at the balance point T_0 . **Noise:** the dominant term is the thermal noise of the bridge's output resistance, $u_n = \sqrt{4k_B T R_{\text{out}} \text{ENBW}}$, which depends only weakly on temperature. For the symmetric case $\alpha_{\text{neg}} = -\alpha_{\text{pos}}$ the two TCRs cancel in R_{out} , so $R_{\text{out}} \approx R$ stays constant and the only temperature dependence of the noise is the slow $\sqrt{T}$. Hence

$$\text{SNR} = \frac{V_o}{u_n} \propto \frac{\Delta T}{\sqrt{T}}.$$

The SNR is zero at T_0 and *increases monotonically* as the temperature moves away from balance, because the numerator climbs linearly while the noise rises only as $\sqrt{T}$.

Additional task: exact SNR. Using the exact $V_o(T)$, $R_{\text{out}} = 2R_n R_p / (R_n + R_p)$ and $u_n = \sqrt{4k_B T R_{\text{out}} \text{ENBW}}$ with $V_B = 5 \text{ V}$, $R = 1 \text{ k}\Omega$, $\text{ENBW} = 1 \text{ kHz}$ and the part-a) α -values gives Fig. 1. The curve rises from $-\infty$ at the 300 K balance point to about 122 dB at 400 K.

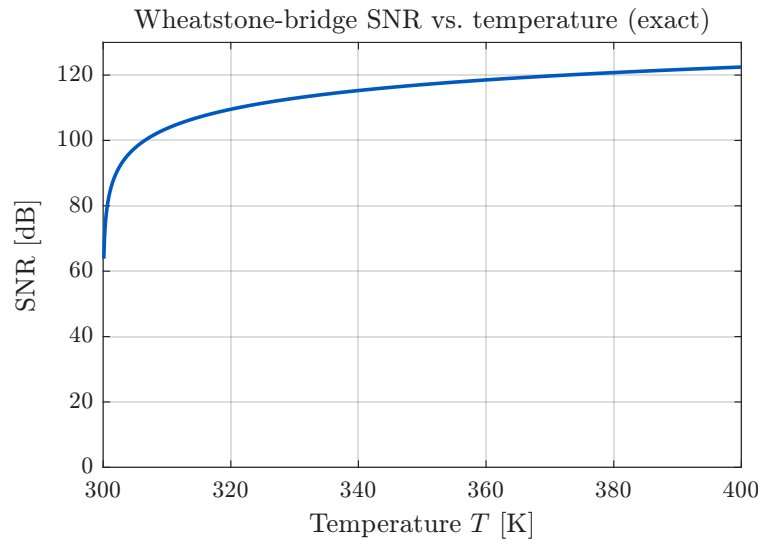


Figure 1: Exact Wheatstone-bridge SNR over 300 K to 400 K. The output (and hence the SNR) vanishes at the 300 K balance point and grows steadily as $|\Delta T|$ increases.

Task 3 c) Measures for sufficient SNR (Additional). The problem is the dead zone near T_0 , where $V_o \rightarrow 0$. Practical fixes:

- **Offset the bridge** (unbalance it deliberately, e.g. trim one arm) so $V_o \neq 0$ at the low end. + removes the dead zone; – costs dynamic range and adds offset drift.
- **Raise the bias V_B :** $\text{SNR} \propto V_B$ while thermal noise is independent of it. + direct gain; – power $\propto V_B^2 \Rightarrow$ self-heating error.
- **Reduce ENBW / average longer:** noise $\propto \sqrt{\text{ENBW}}$. + cheap SNR; – slower measurement (energy $\times$ time trade-off).
- **Higher-TCR materials** (larger $\alpha_{\text{pos}} - \alpha_{\text{neg}}$). + more signal; – worse nonlinearity, needs polynomial correction.

Part C: Photodetectors

Task 2 a) Absorption, thickness and quantum efficiency. (i) Beer–Lambert absorption is $1 - e^{-\alpha d}$, so $d = \ln(\frac{1}{1-A})/\alpha$ with $A = 0.90$ ($\alpha d = \ln 10 = 2.303$) or $A = 0.99$ ($\alpha d = \ln 100 = 4.605$):

	$d_{90\%}$	$d_{99\%}$
600 nm ($\alpha = 4140/\text{cm}$)	5.56 μm	11.1 μm
400 nm ($\alpha = 95\,200/\text{cm}$)	242 nm	484 nm

The strongly absorbed blue light (400 nm) is collected within half a micron, whereas red light (600 nm) needs a $\sim 20\times$ thicker active layer.

(ii) Going from the 90% to the 99% thickness, the absorbed fraction rises $0.90 \rightarrow 0.99$, so the relative change in quantum efficiency is

$$\frac{\Delta\eta}{\eta} = \frac{0.99 - 0.90}{0.90} = +10\%,$$

identical for both wavelengths. Roughly doubling the thickness buys only 10% more efficiency, so the returns diminish quickly.

(iii) With $\eta = (1 - \mathcal{R})(1 - e^{-\alpha d})\zeta$, $\zeta = 1$ and $(1 - e^{-\alpha d}) = 0.99$:

$$\eta(400\text{ nm}) = (1 - 0.485)(0.99) = \mathbf{51.0\%}, \quad \eta(600\text{ nm}) = (1 - 0.354)(0.99) = \mathbf{63.9\%}.$$

Red light reaches a higher external efficiency here purely because its surface reflectivity is lower.

Task 2 b) Integrated photoconductor. $N_D = 1 \times 10^{17}/\text{cm}^3$ ($n = N_D$), $\mu_n = 700 \text{ cm}^2/(\text{V s})$, $\mu_p = 240 \text{ cm}^2/(\text{V s})$, $t = 1 \mu\text{m}$, $W = 10 \mu\text{m}$, $L = 30 \mu\text{m}$, $V = 1.8 \text{ V}$.

(i) Dark current. The well is strongly n -type ($p = n_i^2/n \approx 2 \times 10^3/\text{cm}^3$ is negligible), so $\sigma = nq\mu_n = 11.21 \text{ S/cm}$ and

$$R = \frac{1}{\sigma} \frac{L}{Wt} = \frac{1}{11.21} \cdot \frac{30}{10 \times 1} \times 10^4 \Omega = 2675 \Omega, \quad I_{\text{dark}} = \frac{V}{R} = \mathbf{0.67 \text{ mA}}.$$

(ii) Photocurrent. The photon flux is $\Phi = \phi LW = 10^{16} \times (30 \times 10 \times 10^{-12}) = 3 \times 10^6/\text{s}$. The transit time is $\tau_e = L^2/(\mu_n V) = (3 \times 10^{-3} \text{ cm})^2/(700 \times 1.8) = 7.14 \text{ ns}$, giving a gain $G = (\tau/\tau_e)(1 + \mu_p/\mu_n) = 14.0 \times 1.343 = 18.8$ (Task 2 b iii). Hence

$$i_p = e\eta\Phi G = (1.602 \times 10^{-19})(0.515)(3 \times 10^6)(18.8) = \mathbf{4.65 \text{ pA}}.$$

(The primary photocurrent without recirculation would be $e\eta\Phi = 0.25 \text{ pA}$.)

(iii) Gain and its geometry dependence. Combining the expressions,

$$G = \frac{\tau V (\mu_n + \mu_p)}{L^2} \propto \frac{1}{L^2}, \quad \text{independent of } W.$$

$G = \mathbf{18.8}$. (a) Doubling the length ($2L$) makes τ_e four times longer, so $G \rightarrow G/4 = 4.7$. (b) Doubling the width ($2W$) leaves the gain *unchanged*.

(iv) Photocurrent for those cases. Since $\Phi = \phi LW$ scales with area, $i_p = e\eta\Phi G \propto LW/L^2 = W/L$:

$$(a) \ 2L : i_p \rightarrow \frac{1}{2}i_p \ (\text{area} \times 2, G \times \frac{1}{4}); \quad (b) \ 2W : i_p \rightarrow 2i_p \ (\text{area} \times 2, G \text{ unchanged}).$$

Widening the device is the productive way to collect more photocurrent; lengthening it actually reduces it because the gain loss outpaces the larger collection area.

Part D: IS-MOS Capacitance

Task 4 a) Labelling and characteristic parameters. The curve is a low-frequency MOS C–V on a p -type substrate: at negative V_G holes accumulate ($C = C'_{\text{max}} = C_{ox}$), the capacitance dips through depletion to C'_{min} , and at positive V_G the low-frequency minority response brings it back up. The three red parameters are the accumulation plateau C'_{max} , the flat-band value C'_{FB} (the marked point), and the inversion minimum C'_{min} ; the axes are V_G (horizontal) and C' (vertical), see Fig. 2.

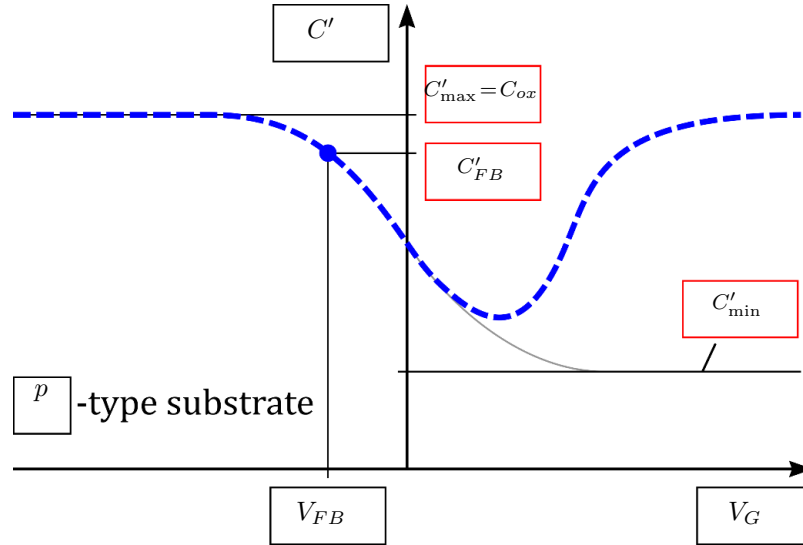


Figure 2: Assignment Figure 3 with the gaps filled in: p -type substrate, y -axis C' and x -axis V_G , the flat-band voltage V_{FB} , and the three red parameters $C'_{\max} = C_{ox}$, C'_{FB} and $C'_{\min}$.

(i) **Values at 300 K** (SiO_2 , $\epsilon_r = 3.9$, $t_{ox} = 30 \text{ nm}$; Si , $\epsilon_r = 11.7$, $N = 3 \times 10^{16}/\text{cm}^3$). All values per unit area:

$$C'_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = 1.15 \times 10^{-3} \text{ F/m}^2 \text{ (115 nF/cm}^2\text{)}.$$

The bulk potential is $\phi_F = \frac{k_B T}{e} \ln \frac{N}{n_i} = 0.375 \text{ V}$, giving a maximum depletion width $W_{\max} = \sqrt{4\epsilon_{Si}\phi_F/(eN)} = 180 \text{ nm}$ and Debye length $L_D = \sqrt{\epsilon_{Si}(k_B T/e)/(eN)} = 23.6 \text{ nm}$. With $C'_{\min} = C_{ox}C_{sc}/(C_{ox} + C_{sc})$ for each series semiconductor capacitance:

Gate / doping	$C'_{\max} = C_{ox}$ [nF/cm ²]	C'_{FB} [nF/cm ²]	$C'_{\min}$ [nF/cm ²]
SiO_2 , $N = 3 \times 10^{16}$	115	91	38.4

(ii) **Si_3N_4 gate** ($\epsilon_r = 7.5$) **and lower doping** $N = 0.5 \times 10^{16}/\text{cm}^3$. The higher- κ oxide nearly doubles C_{ox} , and the lighter doping widens the depletion region ($W_{\max} = 413 \text{ nm}$), lowering $C'_{\min}$:

Gate / doping	$C'_{\max}$ [nF/cm ²]	C'_{FB} [nF/cm ²]	$C'_{\min}$ [nF/cm ²]
Si_3N_4 , $N = 0.5 \times 10^{16}$	221	99	22.6

The capacitance swing grows from $C_{\max} - C_{\min} = 77 \text{ nF/cm}^2$ to 199 nF/cm^2 , a factor of 2.6. With ΔV fixed, the slope $\Delta C/\Delta V$ therefore rises by the same factor, so the change makes a **more sensitive sensor**.

Task 4 b) C–V modifications (low frequency). The flat-band voltage is $V_{FB} = \frac{\phi_M}{e} - \frac{\phi_{Si}}{e} - \frac{Q_{ox} + Q_{ss}}{C_{ox}}$ (EOS adds $E_{REF} - \psi_0 + \chi_{sol}$). Each effect shifts the curve horizontally without changing its shape (Fig. 3):

- (i) Fixed positive Q_{ox} , Q_{ss} lower $V_{FB} \Rightarrow$ shift to the **left** (negative V).
- (ii) A higher semiconductor work function ϕ_{Si} also lowers $V_{FB} \Rightarrow$ shift **left**.
- (iii) Increasing pH makes ψ_0 more negative, and since V_{FB} contains $-\psi_0$ the curve shifts to the **right** ($\leq 59 \text{ mV/pH}$).

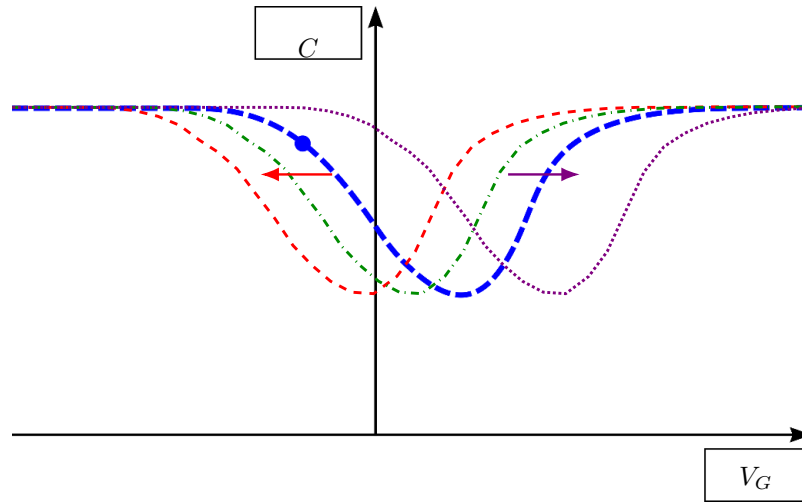


Figure 3: Sketched over assignment Figure 4 (blue = ideal). (i) red, dashed: positive fixed charge shifts left; (ii) green, dash-dot: higher work function shifts left; (iii) violet, dotted: increased pH shifts right.

Task 4 c) EOS-Cap pH sensitivity. In the linear regime the chemical sensitivity is the product of the (shared) electrical slope and the voltage shift per pH, $\Delta C/\Delta \text{pH} = (\Delta C/\Delta V) (\Delta V/\Delta \text{pH})$, with the Nernst limit $\Delta V/\Delta \text{pH} = 2.303 \frac{k_B T}{e} \alpha = 59.5 \text{ mV } \alpha$:

$$\text{A (SiO}_2, \alpha = 0.5) : \frac{\Delta V}{\Delta \text{pH}} = 29.8 \text{ mV/pH}, \quad \text{B (Al}_2\text{O}_3, \alpha = 0.9) : \frac{\Delta V}{\Delta \text{pH}} = 53.6 \text{ mV/pH}.$$

Because both devices share the same slope $\Delta C/\Delta V$, their capacitive sensitivities scale the same way: $(\Delta C/\Delta \text{pH})_B / (\Delta C/\Delta \text{pH})_A = \alpha_B / \alpha_A = 1.8$. The near-Nernstian Al_2O_3 device is $1.8\times$ more sensitive than the sub-Nernstian SiO_2 one.

Task 4 d) Effect of a different semiconductor. The pH response lives at the *oxide–electrolyte* interface (it is set by ψ_0 and the site-dissociation chemistry), so the intrinsic sensitivity in mV/pH does **not** change with the semiconductor. What does change is the baseline: a different work function ϕ_{Si} , bandgap, n_i and permittivity shift V_{FB} and reshape the C–V curve (different $C_{\min}$, depletion width) and alter temperature stability and leakage. In short, the curve moves and rescales, but the pH slope stays the same.

Part E: Introduction to FEM (Unit 5)

1. Three key steps from physics to a numerical discretization.

1. **Mathematical model (strong form).** Capture the physics as a governing PDE plus its boundary and initial conditions: the exact, continuous description.
2. **Weak / variational form.** Multiply by a test function and integrate over the domain, moving derivatives onto the test function. This relaxes the continuity requirement and is what makes the problem amenable to piecewise approximation.
3. **Discretization (Galerkin).** Mesh the domain into finite elements, expand the unknown in a finite set of basis (shape) functions, and assemble the result into an algebraic system $\mathbf{Ku} = \mathbf{f}$ that a computer solves; then post-process and check convergence.

2. Ideal polynomial order of the basis functions. Quadratic (second-order) elements are usually the sweet spot.

- *Lower order* (linear/hat functions) is cheap and robust and represents straight fields exactly, but it only captures curvature crudely, so smooth or high-gradient fields need a very fine mesh to reach the same accuracy.
- *Higher order* (cubic and up) gives excellent accuracy per element and resolves curvature with few elements, but the cost per element climbs, matrices become denser and worse-conditioned, and high orders can oscillate (Runge-type wiggles) near sharp features.

Quadratic elements capture curvature well while keeping the cost and conditioning manageable, which gives the best accuracy per unit effort for typical smooth problems.

3. Qualitative grid spacing for linear (hat) basis functions. Linear hat functions reproduce a straight line *exactly*, so the rule is simple: **use a coarse spacing on straight parts, refine where the curvature is high, and always place a node at every corner/kink** (where the slope is discontinuous).

- (a) **Straight line:** one element (two nodes) is enough, since it is exact.
- (b) **Exponential:** a graded mesh, fine where the function bends sharply and progressively coarser in the flat tail.
- (c) **Mixed curve:** coarse on the straight segments and the triangular peak (with nodes at the joints $x = -4, -2, 2, 4, 6$), but densely spaced across the high-curvature semicircle.

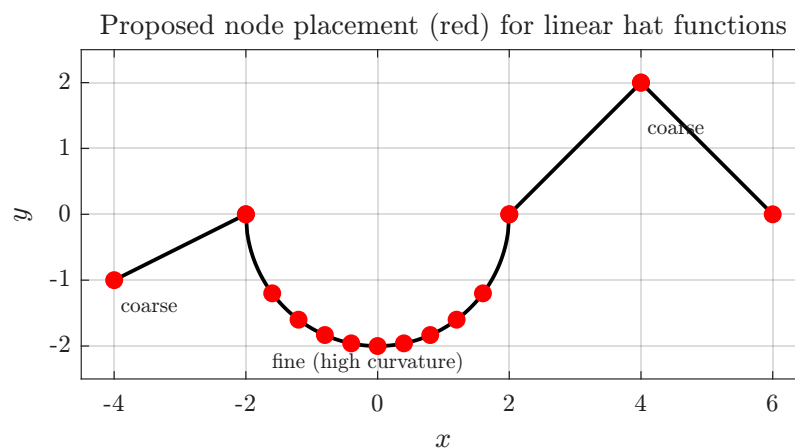


Figure 4: Proposed node placement (red) for curve (c): coarse on the straight pieces, dense across the curved semicircle, with nodes pinned at every kink.